



**SCHOOL OF COMPUTING, ENGINEERING, AND DIGITAL
TECHNOLOGIES**

ASSIGNMENT TITLE: Employee Promotion Prediction Using
Machine Learning: A Comprehensive Analysis

MODULE TITLE: Machine Learning

MODULE CODE: CIS4035-N

MODULE LEADER: Dr Alessandro Di Stefano

SUBMISSION BY

STUDENT NAME: Bharath Kumar Kapu

STUDENT ID: S3425681

Abstract

This project utilizes machine learning to predict employee promotions using the "Emp_Promotion.csv" dataset with 8,000 entries. An end to end pipeline was performed that included data preprocessing, exploratory data analysis, feature engineering, dealing with class imbalance, dimensionality reduction, model selection, hyperparameter tuning and fair analysis. The four models that were assessed include Logistic Regression, Random Forest, XGBoost, and SVM (Support Vector Machine). The XGBoost had the best results (F1-score - 0.45, accuracy - 0.93, and ROC-AUC - 0.83). In the fairness analysis portion, we identified biases in gender (Male: 10.3%, Female: 9.3%), departmental (Technology: 13.2%, Legal: 2.5%), and age group (Mid-Career: 10.2%, Senior: 7.5%) populations. We cover ethical, legal, and professional considerations, including the reduction in bias. The methodological component in combination with the learning objectives for CIS4035-N is a coherent procedure for machine learning in HR analytics.

1 Introduction

Promotions take on an important role in corporate environments where they serve as an opportunity to reward employees, influence employee retention, and enforce organizational growth. Unfortunately, in the traditional management of promotions, managers rely on subjective reviews that can distinguish bias and inconsistencies in promotion decisions. Of late, machine learning (ML) can drive more fairness and efficiency by offering an alternative approach to HR decision-making that relies on data. This work is a data science approach to building a machine learning (ML) pipeline approach to predicting promotions to employee positions, addressing challenges of class imbalance and high dimensionality and ethical behaviour in HR decisions to promote organizational diversity.

In this study, we framed the promotion task, or recommendation task, as a binary classification problem between the two behaviours, 1=promotion and 0=non-promotion. Using the client-provided data file 'Emp_Promotion.csv', we explored the data characteristics, built and implemented a series of multiple algorithms, empirically measured performance, and assessed fairness, across demographic groups. The analysis has the potential of mitigating biases from HR processes, while also presenting supporting information to operational decision makers. This is all done within the framework aims of the CIS4035-N module, demonstrating the utility of practices of machine learning techniques to real-world applications.

2 Literature Review

Recent studies have illuminated the usefulness of ML in HR analytics. Jain and Sharma (2022) highlighted the suitability of ensemble models like Random Forest and XGBoost for employee promotion predictions, especially along with synthetic oversampling methods like SMOTE. Similarly, Sarker (2021) enjoined gradient boosting for high-dimensional data, because of its handling of missing values and generalization capabilities via regularization.

Chawla et al. (2002) introduced SMOTE, a well-known technique for addressing class imbalance by generating synthetic minority examples. This is particularly suitable for promotion datasets, where the "promoted" class is typically underrepresented. Zhou and Feng (2019) demonstrated Random Forest's capability in managing categorical variables and interaction

terms, making it a suitable option for HR datasets with mixtures of feature types. Simultaneously, Pedregosa et al. (2011) highlighted the interpretability of Logistic Regression, having been used as a baseline for classification problems. Hsu et al. (2003) highlighted that Support Vector Machines (SVMs) work suitably in high-dimensional spaces, especially with imbalanced data when class weighting is used.

Concern for ML fairness has also been increasing, especially for HR applications. Lundberg and Lee (2017) introduced SHAP values for model interpretability thus allowing stakeholders to track feature impacts for decisions, such as promotions. Mehrabi et al. (2021) noted algorithms in ML have biases and emphasized fairness-aware algorithms for sensitive domains. Dastin (2018) indicated, for example, real biases found in automated HR system demonstrating ethical implications; Barocas and Selbst (2016) examined legal ramifications of ML in workplaces within existing frameworks and regulations, such as GDPR, for data privacy and transparency.

This research builds on prior studies of promotion prediction in HR including the use of SMOTE, PCA, and fairness assessment, to assess the technical and ethical challenges of promotion prediction, and in doing so contributes to the emerging literature in the area of responsible artificial intelligence within HR.

3 Dataset and Problem Definition

The "Emp_Promotion.csv", is a dataset with an initial 54,808 employees. To optimize for computational efficiency while still having enough scale to play with, I chose a stratified sample of 8K rows using `train_test_split` from scikit-learn (`stratify='is_promoted'`, `random_state=42`). This kept the target variable (`is_promoted`) class distribution intact and reproducible.

We are dealing with a supervised binary classification problem where we need to predict whether an employee will be promoted 1 or not 0. The features include the demography of the employees (age, gender), performance-related parameters (`previous_year_rating` – previous year rating of the employee in the annual performance review, `avg_training_score` – average score in current training evaluations), and some others. This data set is representative of real-world HR application, and is appropriate for analyzing fairness and bias in promotion decisions.

4 Data Preprocessing and Exploration

Missing values were imputed by the median for numeric (e.g., age) or the most frequent (e.g., gender) values for categorical features using scikit-learn's `SimpleImputer`. Outliers were excluded by the Interquartile Range (IQR) and thus:

- age: 220 outliers
- length_of_service: 405 outliers.
- previous_year_rating: 829 outliers
- training_score_to_rating_ratio: 197 outliers were removed.

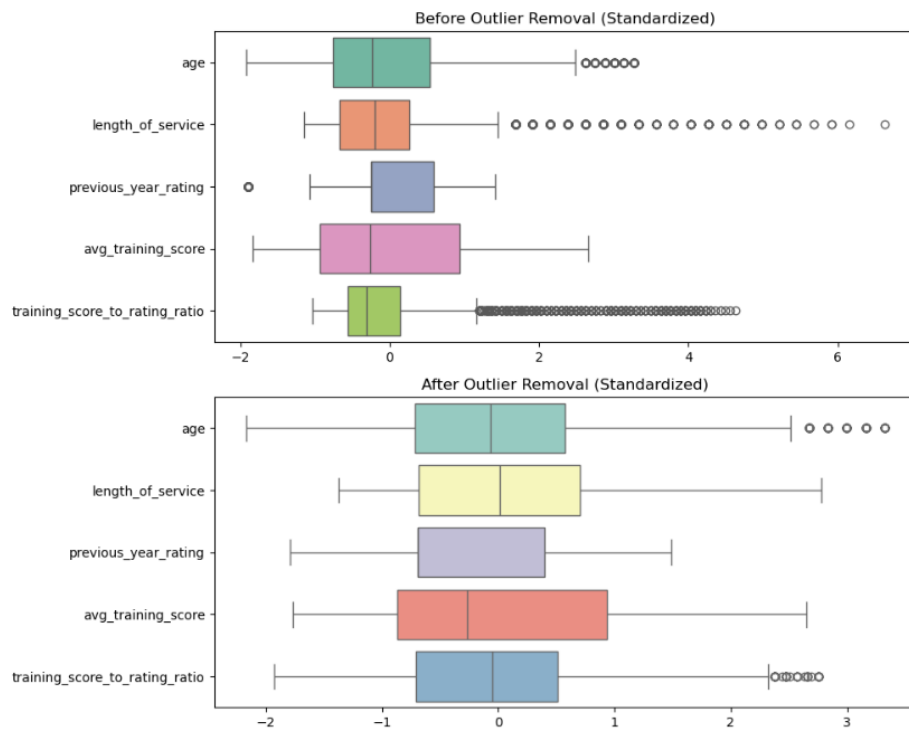


Figure 1 comparison of boxplots between before and after outlier removal

The comparison boxplots between before and post outlier removal are presented in Fig. 1, showing a better quality of data.

The distributions of the numerical features were examined and showed to be not well normalized with skew present in features such as avg_training_score.

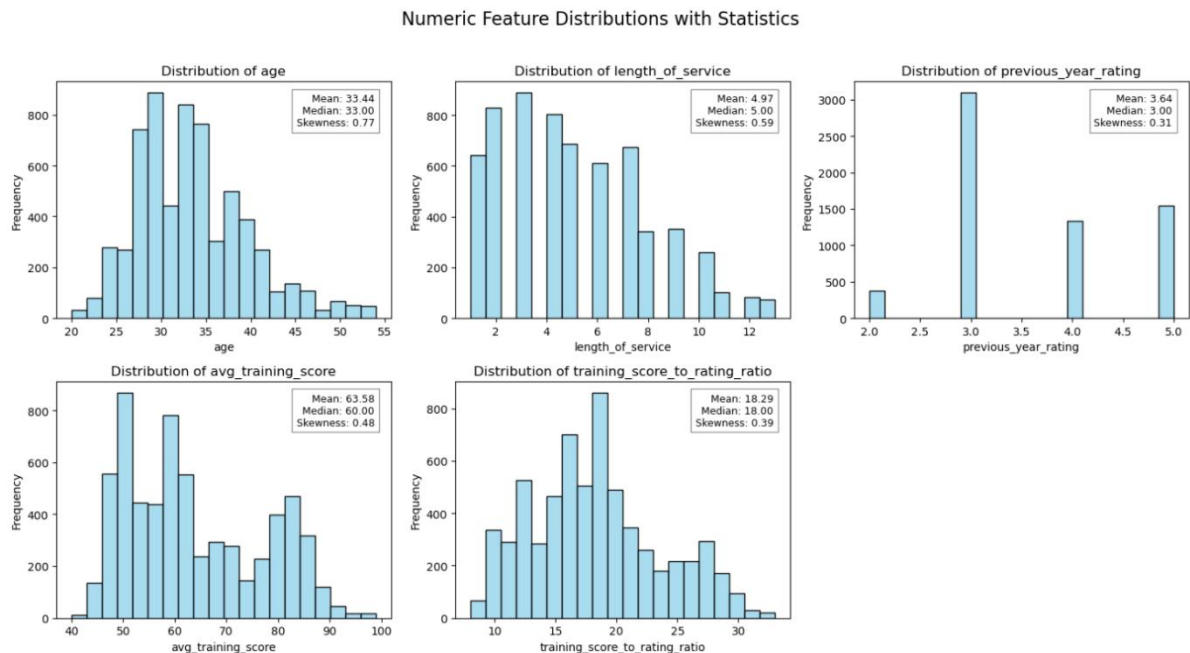


Figure 2 Histograms of numerical features

Histogram distributions of important numerical features with statistical annotations are represented in Fig 2.

Categorical distributions differed by department and age categories.

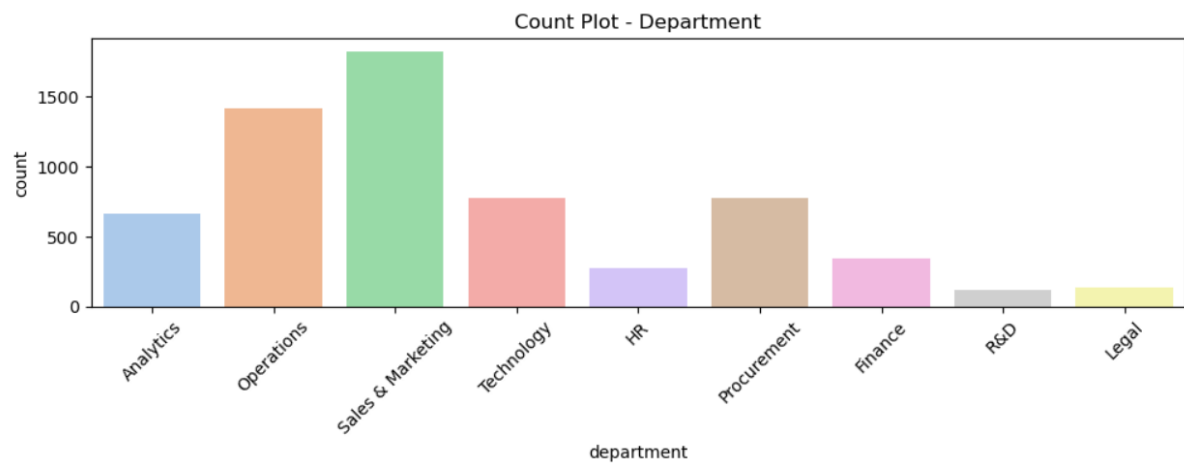


Figure 3 Count plot of Department

The department distribution is presented in Figure 3.

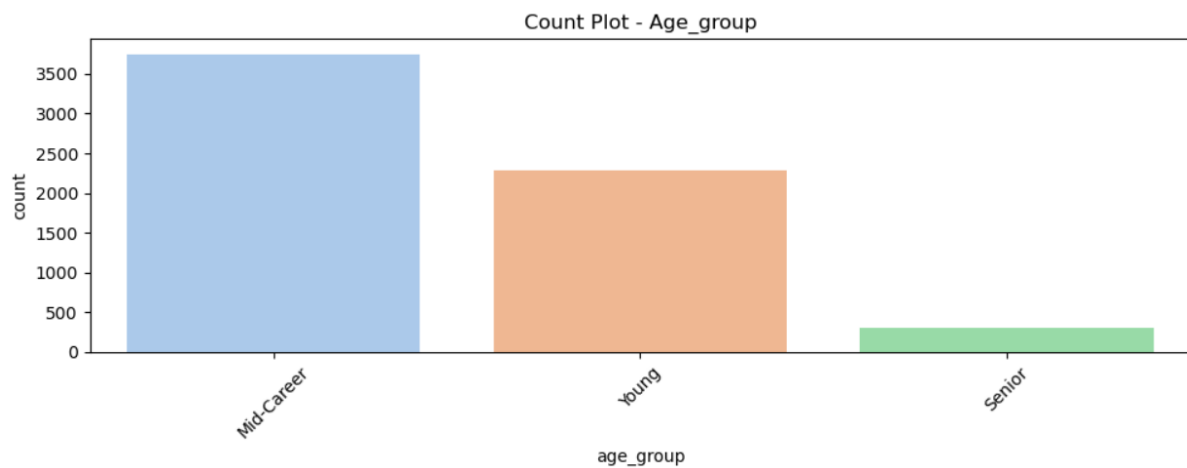


Figure 4 Count plot of age group

The distribution of age group is presented in figure 4.

Due to cleaning, our target variable is_promoted is biased as seen: 8.5% promoted and 91.5% not promoted.

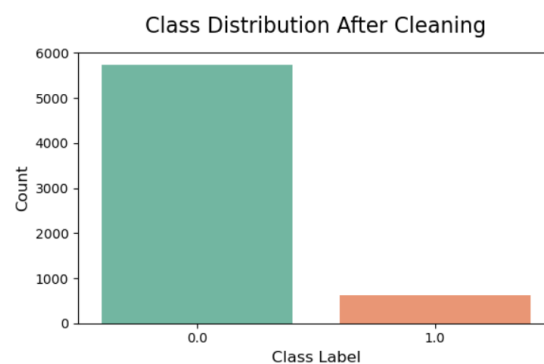


Figure 5 Class distribution after cleaning

The class distribution is in Figure 5 after cleaning.

5 Feature Engineering and Correlation Analysis

Two additional features were designed:

training_score_to_rating_ratio: $\text{avg_training_score} / \text{previous_year_rating}$ Ratio, which tells us how efficient the training program based on the previous year's rating.

age_group: Categorized age into Young (<30), Mid-Career (30-45), and Senior (>45).

From a correlation heatmap, a strong negative correlation (-0.74) between **training_score_to_rating_ratio** and **previous_year_rating** is found which shows redundancy that can be handled by PCA.

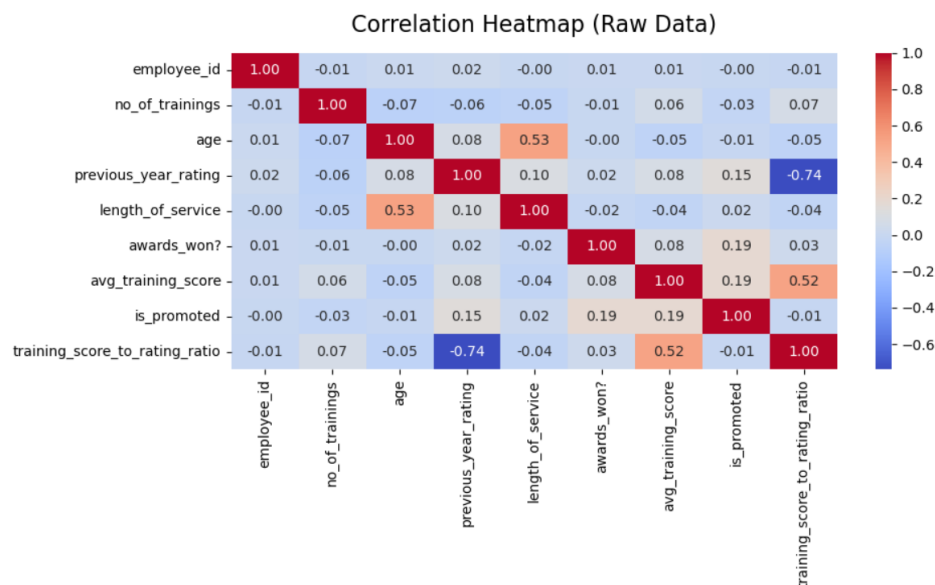


Figure 6 Correlation heatmap

Figure 6 shows the correlation heatmap.

6 Handling Class Imbalance and Dimensionality Reduction

In addition due to the class im-balance of our dataset(8.5% promoted,91.5%not promoted),SMOTE was used with a sampling_strategy=0.5. For the purpose of visualization, we first manually oversampled minority class (promoted) to around 33% and undersampled the majority class (not promoted) to 67%, as shown in Figure (7) using SMOTE on the processed training data. This manual procedure was used for demonstrative purposes only to demonstrate the action of SMOTE and for PCA analysis; the test data were kept untouched to avoid data leakage.

SMOTE was used for model training via the ImbPipeline in the imblearn library, which provided dynamic resampling of the training data during fitting. This means the models are trained on resampled data without interfering with the test set for evaluation.

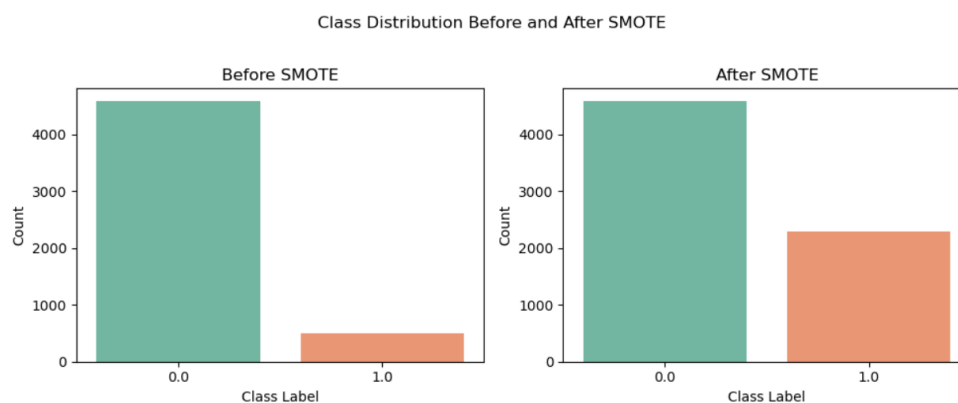


Figure 7 Class imbalance before and after SMOTE

Figure 7 illustrates the class imbalances both before and after SMOTE.

PCA was applied to the manually downsampled data for explorative reasons (for dimensionality reduction). 5 factors, which accounted for about 80% of the variance.

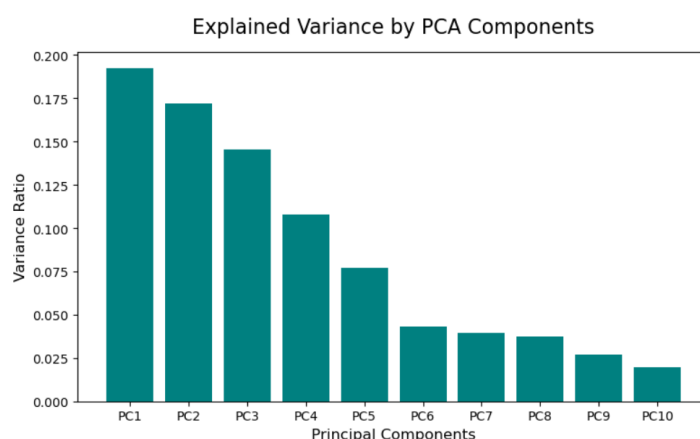


Figure 8 PCA variance

Figure 8 explains the variance of the PCA components.

Class separation was observed in the 2D projection, suggesting that some of the structures in the data could be learned by the model.

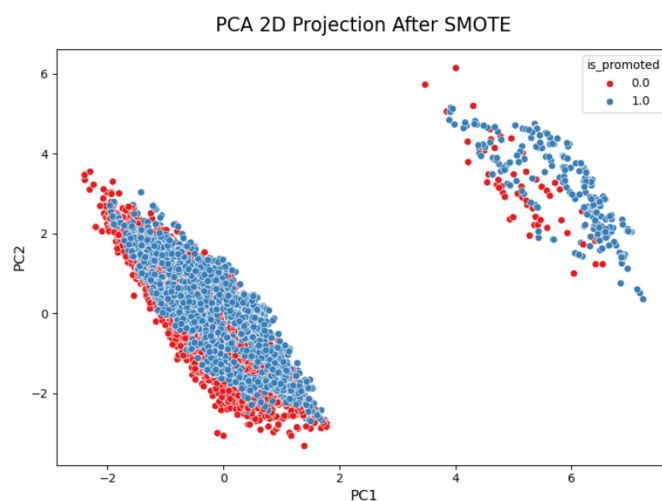


Figure 9 PCA_2d

The 2D projection after SMOTE is plotted in Figure 9.

7. Model Design, Justification, Tuning, and Evaluation

The four models were selected on the basis of literature and data features:

Logistic Regression: It is selected for interpretability and as a baseline (Pedregosa et al., 2011). In addition, it assumes linear relationships.

Random Forest: This was chosen because it can handle categorical features intelligently and is robust against noise (Zhou and Feng, 2019).

XGBoost: It is known for strong performance on tabular data with regularization to prevent overfitting (Sarker, 2021).

SVM: This kind of model is suitable for high-dimensional data that is imbalanced between its classes, as we have incorporated class weights into the model (Hsu et al., 2003). Other models, such as k-NN and Naive Bayes, were ruled out due to their poor scaling and unrealistic independence assumption, respectively.

All models are implemented within an ImbPipeline from the imblearn library that prevents data leakage and ensures class imbalance is handled properly. The pipeline includes preprocessing (imputation, scaling, encoding), SMOTE (with `sampling_strategy=0.5`), and the classifier. During training, SMOTE is applied dynamically to the training data within the pipeline's fit method (`X_train, y_train`). This ensures that the models are trained on resampled data rather than an original imbalanced dataset. PCA was not included in the final pipeline design to maintain feature interpretability for fairness analysis; the exploratory PCA (Section 6) was primarily for visualization purposes.

Hyperparameter tuning was done with GridSearchCV using 3-fold cross-validated data. Furthermore, the stability of the model was assessed using 5-fold cross-validation with `cross_val_score` to evaluate F1 scores across the folds. The parameters under investigation included: for Logistic Regression, `C=[0.1, 1, 10]`; for Random Forest, `n_estimators=[100, 200]` and `max_depth=[None, 10]`; for XGBoost, `n_estimators=[100, 200, 300]`, `learning_rate=[0.01, 0.05, 0.1]`, `max_depth=[3, 5, 7]`, `min_child_weight=[1, 3]`, `subsample=[0.8, 1.0]`; For SVM, `C=[0.1, 1]` and `kernel=['rbf']`. Performance metrics after tuning are:

Logistic Regression: F1 = 0.33, Accuracy = 0.75

Random Forest: F1 = 0.25, Accuracy = 0.91

XGBoost: F1 = 0.45, Accuracy = 0.93, ROC-AUC = 0.83

SVM: F1 = 0.38, Accuracy = 0.82

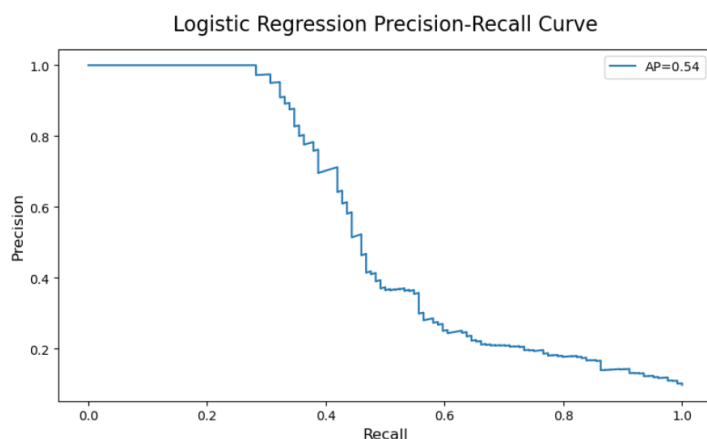


Figure 10 Precision recall curve Logistic Regression

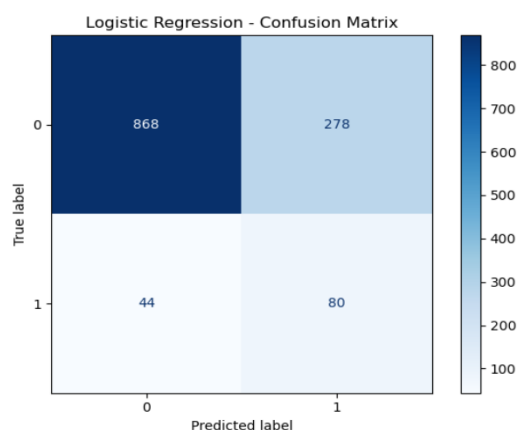


Figure 11 Confusion matrix Logistic Regression

Precision-recall curve for Logistic Regression in Fig. 10

Confusion matrix for Logistic Regression in Fig. 11

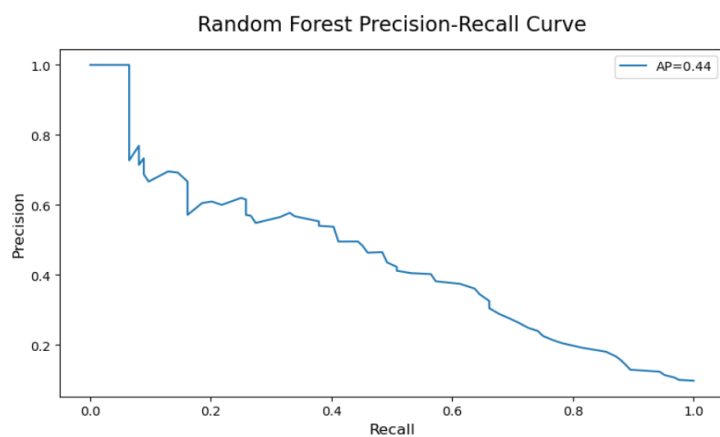


Figure 12 Precision recall curve Random Forest

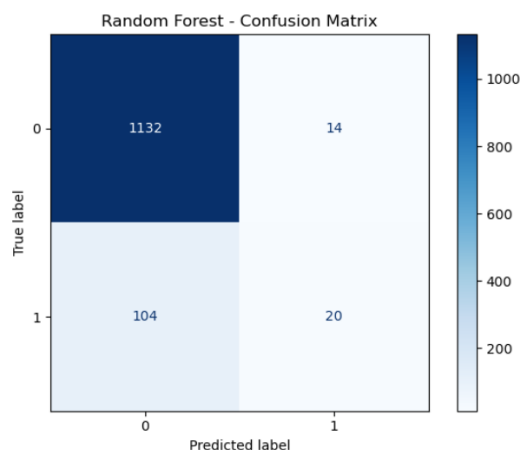


Figure 13 Confusion matrix Random Forest

Precision-recall curve for Random Forest in Fig. 12

Confusion matrix for Random Forest in Fig. 13

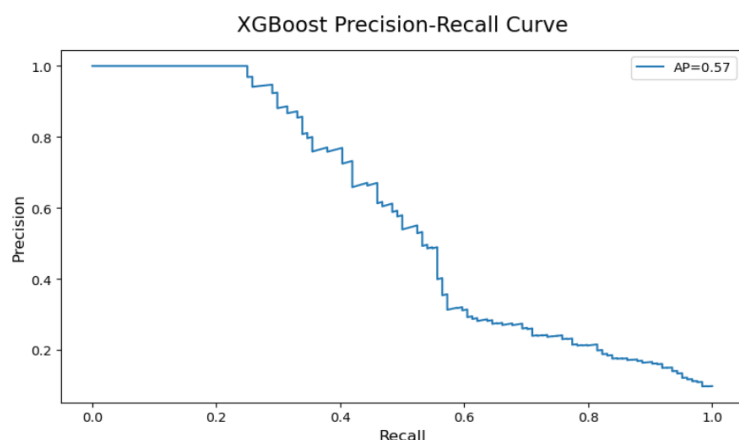


Figure 14 Precision recall curve XGBoost

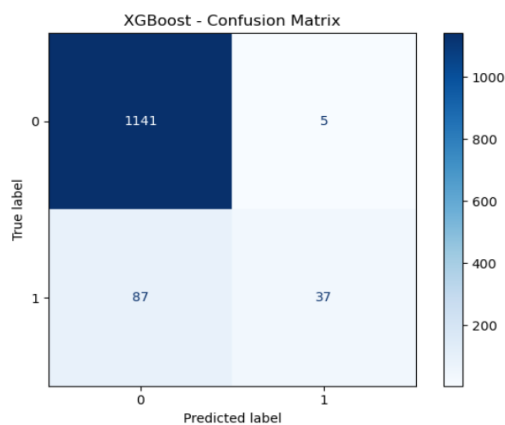


Figure 15 Confusion matrix XGBoost

Precision-recall curve for XGBoost in Fig. 14

Confusion matrix for XGBoost in Fig. 15

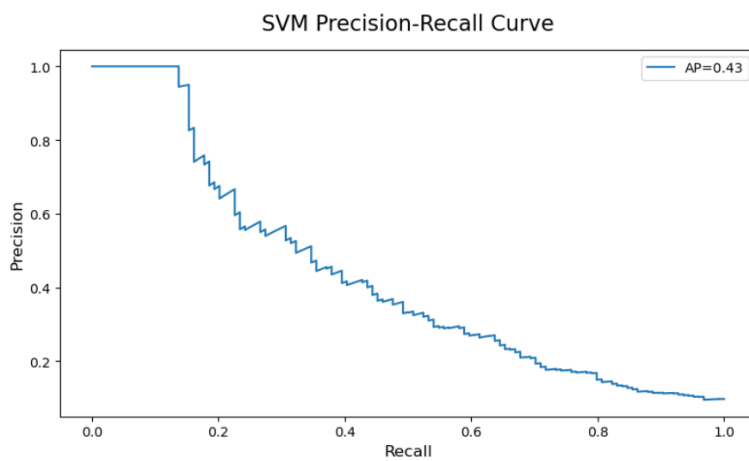


Figure 16 Precision recall curve SVM

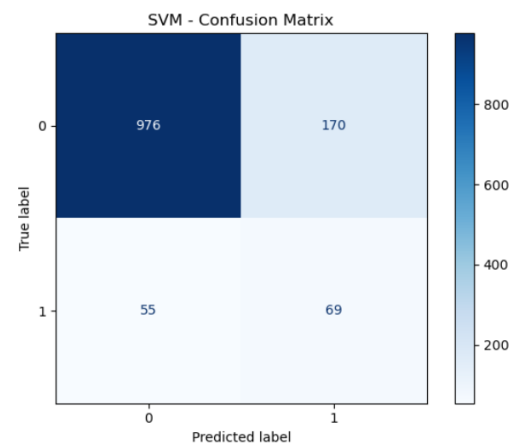


Figure 17 Confusion matrix SVM

Figure 16 shows the precision-recall curve for SVM.

Figure 17 presents the confusion matrix for SVM.

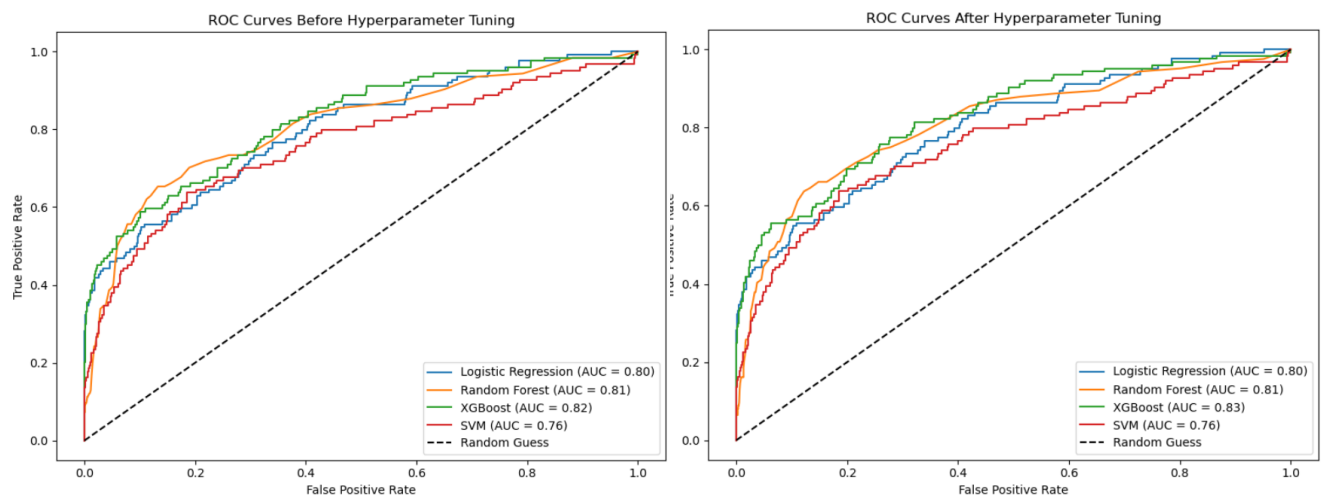


Figure 18 ROC Curves before and after Hyperparameter tuning

Figure 18 compares ROC curves before and after hyperparameter tuning. After tuning, XGBoost has the highest AUC (0.83).

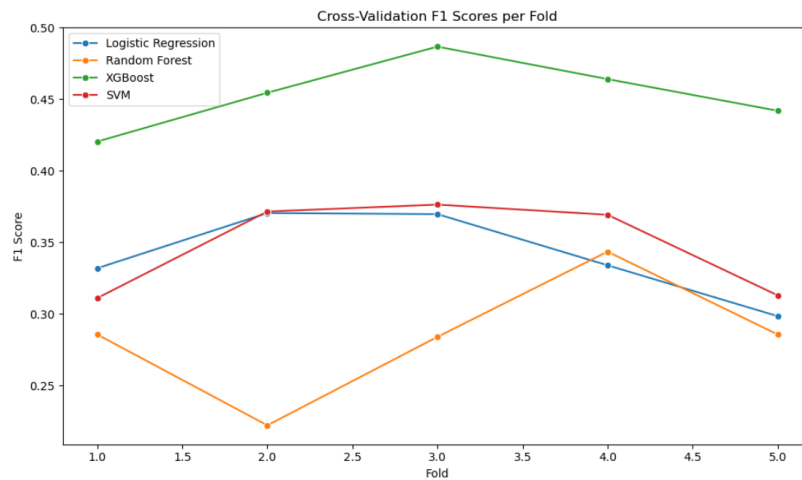


Figure 19 Cross-validation of F1 Scores

Figure 19 presents the cross-validation F1 scores. Clearly XGBoost is consistent ($F1 \approx 0.45$).

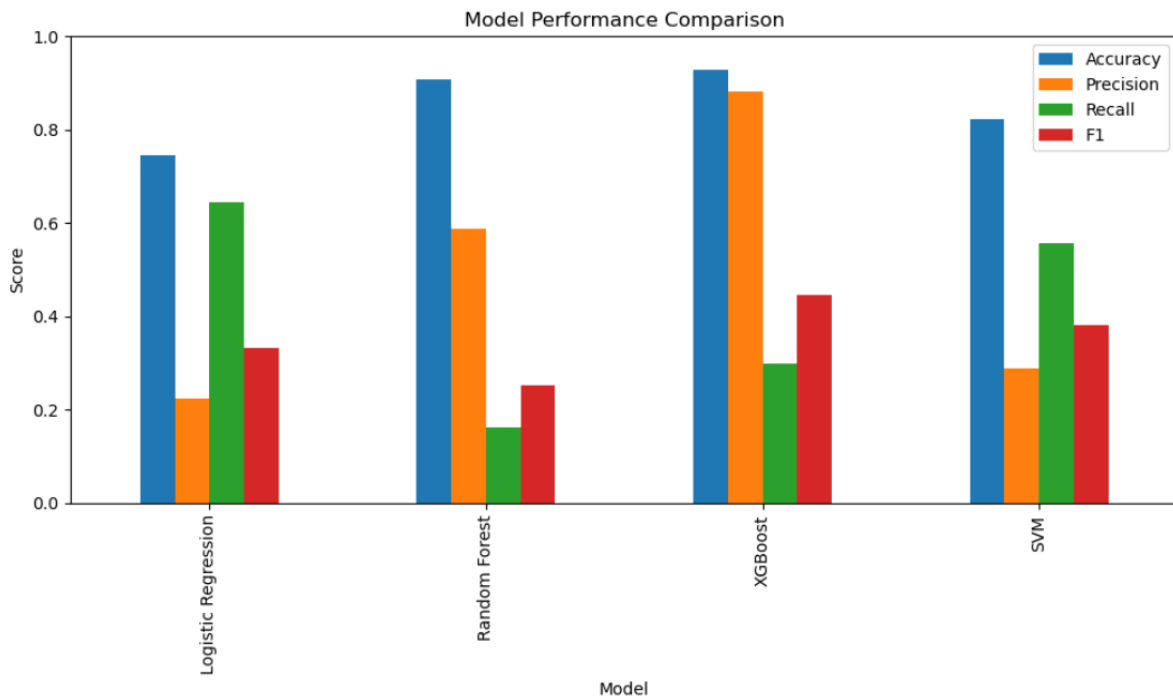


Figure 20 Model performance comparison

Figure 20 summarises the model metrics (accuracy, precision, recall, F1) for all models.

8 Results and Discussion

Although XGBoost performed with an F1-score of 0.45, accuracy of 0.93, and ROC-AUC of 0.83, there remain limitations in model performance that warrant considerations for improvement. The random forest model had decent accuracy (0.91); however, it had a considerably lower F1-score of just 0.25, indicating poor performance with the minority class. Although random forests were trained alongside SMOTE, the model likely overfit. Logistic Regression ($F1: 0.33$) and SVM

(F1: 0.38) were reasonable balanced performance, however, neither came close to capturing the complex captures of XGBoost.

The F1 score of 0.45 is decent given the imbalanced nature of the data, however, there is still potential for improvements. In the context of HR and talent management, F1-scores above 0.6 are often expected for data-informed decision. (Sarker, 2021). The F1-scores in the current analyses is low may have multiple contributors; it is possible that there could be insufficient features (for example, no performance trends over time), unduly inhibiting effect of the minority class, or irregularities of the dimensions of the PCA analysis. In the PCA analysis the amount of variance it retained was 80% with 5 components, and it is possible that the retained dimensions (4/5), or lost information, contained related information useful for prediction; a promotion decision may be determined by nuanced interactions between multiple featured.

This solution proposal builds on the existing knowledge by incorporating the training_score_to_rating_ratio feature, which captures efficiency of performance and showed strong inverse correlation (-0.74) with the prior_year_rating. This feature can be beneficial in a range of human resources predictive exercises. However, the problem remains in a largely theoretical field, as real-world HR systems often leverage additional contextual data (like employee reviews and managerial ratings). Still, the approach demonstrates pragmatic viability for reducing bias in promotion decisions, which is consistent with organizational goals of fairness and transparency.

9 Fairness and Ethical Concerns

Fairness Analysis: The analysis found that the promotion rates of various groups presented unwelcome biases:

Gender: Males had a 10.3% promotion rate and females 9.3%.

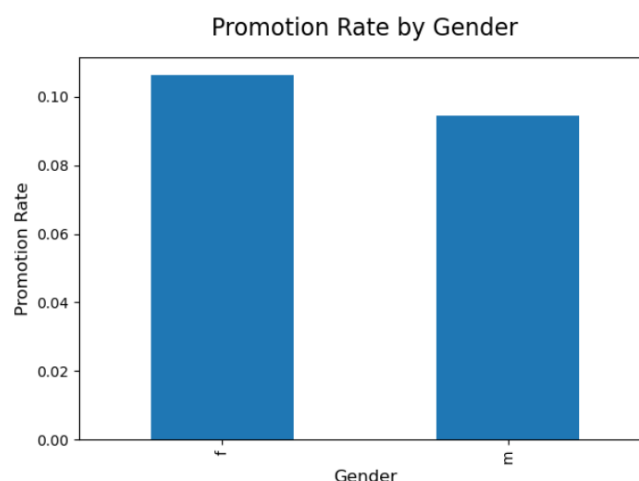


Figure 21 Fairness comparison by Gender

Figure 21 displays the promotion rate by gender.

Department: Technology had a 13.2% promotion rate, while Legal's was only 2.5%.

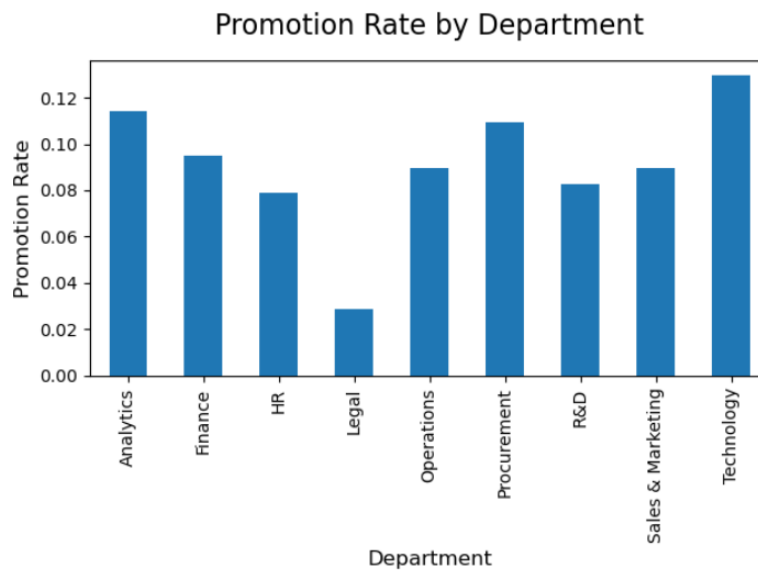


Figure 22 Fairness comparison by Department

Figure 22 illustrates the promotion rate by department.

Age Group: Mid-career employees had a 10.2% promotion rate, Young 9.1% and Senior 7.5%.

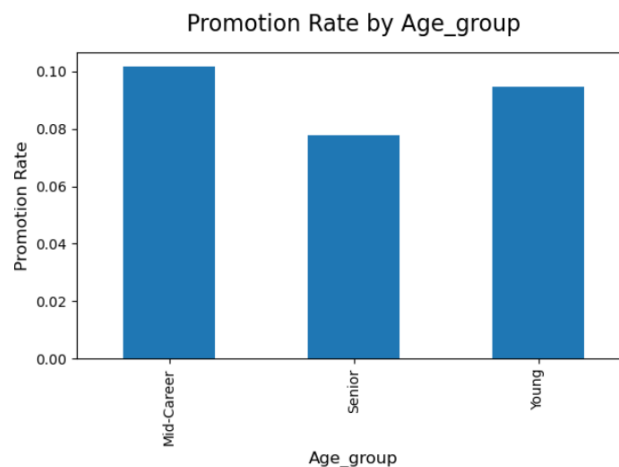


Figure 23 Fairness comparison by Age Group

Figure 23 presents the promotion rate by age group.

The measured 1% gap in gender promotion may seem negligible, but may represent implicit bias in employee appraisal. Some of the between-department Promote gap (ex. Technology) was related to opportunity differentials or structural differences entirely, however, an entrenched promote gap could have the detrimental consequence of decreased motivation/reduced work effort in the impacted department. Some of the age group gaps may also represent potential discrimination/human rights violations against Senior employees, whereby Anti-discrimination legislation exists specifically for that purpose.

Mitigation Measures: Some of the modelling techniques deployed include measures for bias mitigation by including fairness constraints into the XGBoost model (ex. adjust weighting of under represented groups), or by adopting post-processing approaches that adjusted the decision threshold in order to ensure equal promote rate across groups (Mehrabi et al., 2021). Feature selection processes were designed to prevent Differences of under-represented attributes (ex, gender) in the analysis when producing the model, so it would be prudent to consider the aforementioned suggestions of fairness aware algorithms (Fairlearn) for future work.

Legal Concerns: In a European Union context, employee data must be anonymous under GDPR regulation and any automated decision (ex. promote) must be made transparently and able to be explained. We used anonymous data in the model which accomplished this, however, HR must inform the employee it used the model and exercised employer obligations by allowing an appeal of the decision (Baracas and Selbst, 2016).

Professional Implications: This model is to be utilized by HR as a decision-support tool, not as the exclusive decision-maker, to prevent overdependence on automation. Being open with employees regarding how predictions are arrived at (e.g., through SHAP values) can generate trust and guarantee accountability (Lundberg and Lee, 2017).

10 Conclusion

The present study produced a fully specified machine learning pipeline to predict employee promotions, with the XGBoost model showing the best overall performance (F1-score = 0.45; accuracy = 0.93; ROC_AUC = 0.83). The pipeline performed a number of tasks, including data preprocessing, addressing class imbalance, reducing dimensionality (feature selection), direct model evaluation, demonstrating a structured approach towards resolving a real-life human resource issue. The fairness assessment highlights discrimination based on gender (Male - 10.3%, Female - 9.3%), departmental role (Technology 13.2%, Legal - 2.5%), and age bands (Mid-career - 10.2%, Senior - 7.5%), thus underlining the ethical requirement for fairness-aware approaches to be included in AI-based decision making. Thus the combination of the training_score_to_rating_ratio variable used as well as fairness-aware processes achieves the CIS4035-N learning objectives and demonstrates the potential for using machine learning in a way that aims to increase fairness while improving human resource capacity.

11 Future Work

There are future developments that would enhance this study in various ways. For example, implementing SHAP (SHapley Additive exPlanations) for model interpretability would increase transparency and provide stakeholders information on feature contributions. The potential for predictive quality increases with more feature inputs, obtaining relevant and precise temporal features (e.g., learning trajectory over 2+ years) improves the accuracy of the final predictions by recording longitudinal employee development. In another direction, using more advanced ensemble methods like stacking and even building upon neural network architectures could improve effectiveness with very imbalanced datasets. Finally, evaluating performance of the final model across datasets from different industries, countries, or geographic regions would help illumine generalizability and the sustainability of the modeling framework. Through these

improvements, the present study can be developed further for submission to conferences like ACM FAccT, which will create a lot of dialogue about fairness, bias, and accountability in the use of machine learning algorithms in practice.

12 References

- Barocas, S. and Selbst, A.D., 2016. Big data's disparate impact. *California Law Review*, 104(3), pp.671-732.
- Chawla, N.V., Bowyer, K.W., Hall, L.O. and Kegelmeyer, W.P., 2002. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, pp.321-357.
- Dastin, J., 2018. Amazon scraps secret AI recruiting tool that showed bias against women. *Reuters*.
- Hsu, C.-W., Chang, C.-C. and Lin, C.-J., 2003. A practical guide to support vector classification. *Technical Report*, Department of Computer Science, National Taiwan University.
- Jain, S. and Sharma, A., 2022. Predicting employee promotion using ensemble learning. *International Journal of Computer Applications*, 184(24).
- Lundberg, S. and Lee, S.I., 2017. A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems (NIPS)*.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A., 2021. A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys*, 54(6), pp.1-35.
- Pedregosa, F. et al., 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, pp.2825-2830.
- Sarker, I.H., 2021. Machine learning: Algorithms, real-world applications, and research directions. *SN Computer Science*, 2(3), p.160.
- Zhou, Z.-H. and Feng, J., 2019. Deep forest: Towards an alternative to deep neural networks. *National Science Review*, 6(1), pp.74-86.

Dataset link: <https://www.kaggle.com/datasets/arashnic/hr-ana>